

Source-Addressed Stabilizer Descent in Five-Token Return Dynamics

Orbit Incidence and Capacity-Isolated Order Breaking

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This paper (Paper XXXIII of the RIME program) studies intrinsic raw spectator-safe return dynamics in the full lane-partition stabilizer. It imports the normalized partial-return system and lane-stabilizer boundary of Paper XXXII, but no typed transfer, projectability, recursive-return, credit-settlement, or reset-bound conclusion.

Abstract

Problem. Paper XXXII proves that every permutation preserving the arithmetic-lane partition satisfies

$$\text{LaneAdj}_\Delta \implies \text{SafeHit}_a \implies \text{SameLane}_\Delta .$$

Outside the normalizer of the universal return cycle, however, the branch permutation does not induce a deterministic map on return orbits. Separate existential orbit edges also cannot be composed without retaining the same intermediate state.

Approach. For universal-return orbits O, O' , we introduce the source-addressed incidence

$$\mathcal{I}_a(O, O') = a_\#(O) \cap O' .$$

This set of concrete states gives an exact factorization of every stabilizer return. Choosing one representative per orbit further compresses the incidence to a finite cyclic phase set while preserving the intermediate witness.

Results. The source-addressed factorization is exact for every branch permutation. The paper then disproves the natural full-stabilizer promotion claim

$$\text{Break}_\Delta(a) \neq \emptyset \implies \text{SafeHit}_a = \text{SameLane}_\Delta .$$

For every $g \geq 2$, take $n = 5g$ and $\Delta = g$. The punctured lane has capacity four and every ordinary lane has capacity five. A branch may break cyclic order only on the punctured lane while fixing every ordinary lane. A complete five-token context on an ordinary lane cannot enter the breaking lane through any guarded return. The same-lane nonedge contexts on ordinary lanes therefore form a forward-invariant, target-disjoint family.

Boundary. The obstruction does not prove that the break set itself fails to determine Safe-Hit; that stronger non-descent claim requires matched branches with equal retained summaries and different hit predicates. The remaining positive problem is a same-witness, lane-type-sensitive mobility theorem.

Keywords. synchronizing automata; single-defect automata; partial dynamics; orbit incidence; same-witness descent; lane stabilizers; guarded reachability

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled circular carrier
$p(q) = q + 1$	fixed full cycle
Δ	oriented kernel separation
$g = \gcd(n, \Delta)$	number of arithmetic lanes before puncturing
$\ell = n/g$	full lane length
$E = Q \setminus \{0\}$	normalized branch carrier
$\widehat{\Sigma} = \text{Conf}_{2,3}(E)$	two marked tokens and three spectators
ψ_Δ	punctured rotation on E
$H_\Delta = \langle \psi_\Delta \rangle$	globally enabled cyclic return subgroup
$\mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta$	universal-return orbit space
$F_r^{(0)}$	identity-branch partial return with label r
$F_r^{(a)}$	branch- a partial return
$\mathcal{C}_\Delta$	arithmetic-lane partition of E
$\text{Stab}(\mathcal{C}_\Delta)$	full lane-partition stabilizer
$\mathcal{I}_a(O, O')$	concrete incidence $a_\#(O) \cap O'$
$\Theta_a(O, O')$	cyclic phases realizing that incidence
$\text{Break}_\Delta(a)$	lanes on which a is not a cycle-graph isomorphism

1 Introduction

Papers XXIX–XXXII successively replace a historical transfer interface by an intrinsic rank-five raw dynamical system. Paper XXIX identifies the exact spectator-safe quotient and classifies the canonical frontier. Paper XXX extracts the universal return group of a general binary-kernel defect. Paper XXXI normalizes all partial returns into branch coordinates. Paper XXXII classifies the complete single-lane branch space and isolates the full lane-partition stabilizer as the maximal remaining lane-based regime [4–7].

The inherited raw system has state space

$$\widehat{\Sigma} = \text{Conf}_{2,3}(E), \quad (1.1)$$

where two tokens are marked, three are spectators, and every other point is a hole. The globally enabled punctured rotation

$$\psi_\Delta \quad (1.2)$$

generates a cyclic group

$$H_\Delta = \langle \psi_\Delta \rangle \quad (1.3)$$

on this state space.

For a normalizer branch a , the action $a_\#$ descends to a deterministic permutation of

$$\mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta. \quad (1.4)$$

The normalized dynamics is then a deterministic orbit twist followed by an identity-branch relation. The full stabilizer removes exactly this convenience. A branch may carry every arithmetic lane to a lane while failing

$$aH_\Delta a^{-1} = H_\Delta. \quad (1.5)$$

There is then no branch map on $\mathcal{Z}_\Delta$.

The first task of this paper is to identify the exact replacement. The replacement cannot be only a Boolean relation between orbits. If one existential witness establishes that the branch

reaches an intermediate orbit and another establishes that an identity return leaves that orbit, the two statements need not compose. The intermediate state must be shared.

This leads to the source-addressed incidence

$$\mathcal{J}_a(O, O') = a_{\#}(O) \cap O'. \quad (1.6)$$

The object is small enough to be parameterized by one cyclic phase and rich enough to preserve the same witness.

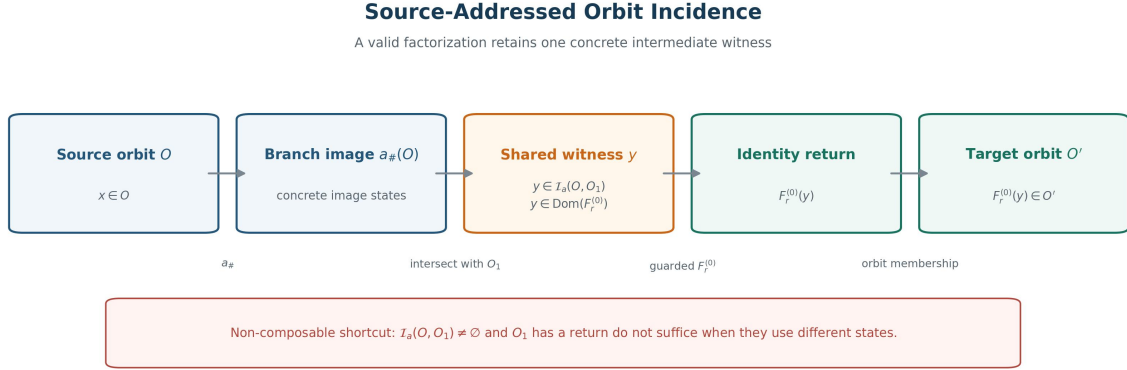


Figure 1. Source-addressed orbit incidence retains one concrete intermediate witness through the branch image and guarded identity return. Separate existential orbit edges do not compose when they use different states.

The second task is to determine whether one order-breaking lane already promotes every same-lane context. It does not. The obstruction is geometric, not merely algebraic: a breaking lane may be too short to receive the full five-token context. This yields an all- g family with a nonempty break set but a proper intermediate Safe-Hit set.

The theorem spine is:

1. Section 2 freezes the normalized branch system and the stabilizer sandwich inherited from Paper XXXII.
2. Section 3 proves exact source-addressed orbit descent.
3. Section 4 compresses concrete incidence to cyclic phase data.
4. Section 5 constructs the all- g capacity-isolation obstruction.
5. Section 6 states the refined same-witness mobility problem.

The work remains intrinsic and raw. It neither rebuilds typed transfer authority nor imports downstream success into the return system.

1.1 Related Work and Novelty Boundary

The Černý conjecture and the wider theory of synchronizing automata concern short reset words [3, 14, 15]. Circular, almost-group, and completely reachable automata provide adjacent structural settings [1, 2, 8, 9, 11, 12]. This paper proves no reset bound and no complete-reachability characterization.

Subset synchronization, careful synchronization, monotonicity, and dynamic word constraints study related partial or order-sensitive behavior [10, 13, 16]. The present state carries

a source-marked pair and three source-marked spectators. One guarded path must keep all five lineages injective until the declared terminal incidence.

Within the RIME line, Papers XXIX–XXXII establish the raw rank-five frontier, the universal return group, branch-normalized partial returns, and the single-lane permutation dichotomy [4–7]. The new claims here are an exact same-witness descent object for the full lane stabilizer and an all- g obstruction to promotion by break-set nonemptiness.

2 The Inherited Stabilizer System

Fix

$$n \geq 6, \quad 1 \leq \Delta \leq n-1, \quad g = \gcd(n, \Delta), \quad \ell = \frac{n}{g}. \quad (2.1)$$

Put

$$E = \mathbb{Z}/n\mathbb{Z} \setminus \{0\}. \quad (2.2)$$

The punctured rotation ψ_Δ has one distinguished cycle C_* and $g-1$ ordinary cycles:

$$\mathcal{C}_\Delta = \{C_*, C_1, \dots, C_{g-1}\}, \quad (2.3)$$

with

$$|C_*| = \ell - 1, \quad |C_j| = \ell. \quad (2.4)$$

Let

$$a \in \text{Sym}(E) \quad (2.5)$$

be an arbitrary branch permutation.

For each label $r \in \mathbb{Z}/n\mathbb{Z}$, Paper XXXI defines the identity-branch partial point map

$$\phi_r^{(0)} = \varepsilon_\Delta p^r, \quad (2.6)$$

and Paper XXXII extends it to the branch map

$$\phi_r^{(a)} = \varepsilon_\Delta p^r a. \quad (2.7)$$

Push-forward to five-token states gives partial maps

$$F_r^{(0)} \quad \text{and} \quad F_r^{(a)} = F_r^{(0)} \circ a_\#. \quad (2.8)$$

The domain guard is injectivity on the complete five-point support. The target is the inherited branch-normalized marked-pair target. Safe-Hit means reachability of that target through the declared partial maps.

Sections 3 and 4 retain the full scope $a \in \text{Sym}(E)$. Beginning in Section 5, the paper restricts to

$$a \in \text{Stab}(\mathcal{C}_\Delta).$$

Such a branch fixes the unique short lane setwise and permutes the ordinary lanes, with arbitrary internal permutations. On this restricted scope, Paper XXXII proves the full-stabilizer sandwich

$$\boxed{\text{LaneAdj}_\Delta(x) \implies \text{SafeHit}_a(x) \implies \text{SameLane}_\Delta(x)}. \quad (2.9)$$

The restriction used for the obstruction theorem does not narrow the algebraic factorization proved in Sections 3 and 4.

3 Source-Addressed Orbit Descent

Let

$$\pi_H : \widehat{\Sigma} \longrightarrow \mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta \quad (3.1)$$

be the universal-return orbit map. For orbits O, O' , define

$$\boxed{J_a(O, O') := a_\#(O) \cap O'}. \quad (3.2)$$

The values of J_a are concrete normalized five-token states. Define the direct branch edge by

$$O \xrightarrow{r,a} O' \quad (3.3)$$

when there exists $x \in O$ such that

$$x \in \text{Dom}(F_r^{(a)}) \quad \text{and} \quad F_r^{(a)}(x) \in O'. \quad (3.4)$$

Proposition 3.1 (exact source-addressed stabilizer descent).

For every branch permutation a , label r , and orbits O, O' ,

$$\boxed{O \xrightarrow{r,a} O' \iff \exists O_1 \in \mathcal{Z}_\Delta \exists y \in J_a(O, O_1) \cap \text{Dom}(F_r^{(0)}) : F_r^{(0)}(y) \in O'}. \quad (3.5)$$

Proof. Suppose $x \in O$ realizes the direct branch edge. Set

$$y = a_\#x \quad (3.6)$$

and let $O_1 = \pi_H(y)$. Then

$$y \in a_\#(O) \cap O_1 = J_a(O, O_1). \quad (3.7)$$

By (2.8),

$$F_r^{(0)}(y) = F_r^{(a)}(x) \in O'. \quad (3.8)$$

Conversely, let y satisfy the right side. Incidence supplies $x \in O$ with $y = a_\#x$. The domain and image clauses then give (3.4) by (2.8).

The concrete state y is essential. In general,

$$J_a(O, O_1) \neq \emptyset \quad (3.9)$$

and

$$\exists z \in O_1 \cap \text{Dom}(F_r^{(0)}) : F_r^{(0)}(z) \in O' \quad (3.10)$$

do not imply (3.5), because the incidence witness and the return witness may be different.

4 Cyclic Phase Compression

Let

$$M = \text{ord}(\psi_\Delta). \quad (4.1)$$

Choose one representative x_O for every $O \in \mathcal{Z}_\Delta$. Define the phase-incidence set

$$\Theta_a(O, O_1) = \{t \in \mathbb{Z}/M\mathbb{Z} : \pi_H(a_\#\psi_\Delta^t x_O) = O_1\}. \quad (4.2)$$

Corollary 4.1 (cyclic phase incidence).

For every O, O_1 ,

$$\mathcal{I}_a(O, O_1) = \{a_{\#}\psi_{\Delta, \#}^t x_O : t \in \Theta_a(O, O_1)\}. \quad (4.3)$$

Moreover,

$$O \xrightarrow{r, a} O' \iff \exists t \in \mathbb{Z}/M\mathbb{Z} : \begin{cases} a_{\#}\psi_{\Delta, \#}^t x_O \in \text{Dom}(F_r^{(0)}), \\ \pi_H(F_r^{(0)}(a_{\#}\psi_{\Delta, \#}^t x_O)) = O'. \end{cases} \quad (4.4)$$

Proof. Every point of O has the form $\psi_{\Delta, \#}^t x_O$. Applying $a_{\#}$ gives (4.3), and substitution into Proposition 3.1 gives (4.4).

Replacing x_O by another representative translates the phase coordinate t . The concrete relation represented by (4.4) is unchanged.

Corollary 4.1 is a finite parametrization of source-addressed incidence. It is not a deterministic action of a on orbit space and not a Safe-Hit classification.

5 Capacity-Isolated Order Breaking

For a lane $C \in \mathcal{C}_{\Delta}$, say that a breaks C if

$$a|_C : C \longrightarrow a(C) \quad (5.1)$$

is not a cycle-graph isomorphism. Put

$$\text{Break}_{\Delta}(a) = \{C \in \mathcal{C}_{\Delta} : a \text{ breaks } C\}. \quad (5.2)$$

The stabilizer sandwich suggests the possible strengthening

$$\text{Break}_{\Delta}(a) \neq \emptyset \stackrel{?}{\implies} \text{SafeHit}_a = \text{SameLane}_{\Delta}. \quad (5.3)$$

The next theorem disproves (5.3) uniformly.

Theorem 5.1 (capacity-isolated break obstruction).

For every integer $g \geq 2$, set

$$n = 5g, \quad \Delta = g. \quad (5.4)$$

There is a branch

$$a_g \in \text{Stab}(\mathcal{C}_{\Delta}) \quad (5.5)$$

such that

$$\text{Break}_{\Delta}(a_g) = \{C_{*}\} \quad (5.6)$$

but

$$\text{SafeHit}_{a_g} \subsetneq \text{SameLane}_{\Delta}. \quad (5.7)$$

For every ordinary lane C_j , each state occupying all five points of C_j with a marked nonedge belongs to

$$\text{SameLane}_{\Delta} \setminus \text{SafeHit}_{a_g}. \quad (5.8)$$

Proof. Under (5.4), the punctured lane has length

$$|C_*| = \ell - 1 = 4, \quad (5.9)$$

and every ordinary lane has length

$$|C_j| = \ell = 5. \quad (5.10)$$

Choose a_g to be an adjacent transposition on the punctured four-cycle and the identity on all ordinary lanes. An adjacent transposition is not a cycle-graph automorphism of a four-cycle, so (5.6) holds.

Let $\mathcal{U}$ be the set of states whose five occupied points fill one ordinary lane and whose marked pair is a cycle nonedge in that lane. Every state in $\mathcal{U}$ is same-lane.

Consider an enabled return from $x \in \mathcal{U}$. Since a_g is the identity on the supporting ordinary lane, the branch step leaves the complete five-point support there. A rotation can carry that full lane either to another ordinary lane or to the five-point unpunctured lane through $0, \Delta, 2\Delta, 3\Delta, 4\Delta$.

The second case is not guarded: the rotated support contains both collision coordinates 0 and Δ . Equivalently, the four-point punctured lane cannot receive a complete five-token context. Hence every enabled return carries x bijectively to a complete ordinary lane.

On ordinary lanes the branch is identity, and the rotational transport is a cycle-graph isomorphism. It therefore preserves marked nonedges. Thus

$$\mathcal{U} \text{ is forward invariant.} \quad (5.11)$$

The normalized branch target on an ordinary lane requires the marked pair to be a Δ -edge of that lane. No state of $\mathcal{U}$ is a target. Therefore $\mathcal{U}$ is target-disjoint, and every state in $\mathcal{U}$ fails Safe-Hit. This proves (5.7) and (5.8).

Corollary 5.2 (break nonemptiness is insufficient).

In the full lane-partition stabilizer,

$$\boxed{\text{Break}_\Delta(a) \neq \emptyset \not\Rightarrow \text{SafeHit}_a = \text{SameLane}_\Delta.} \quad (5.12)$$

The failure is a same-witness mobility failure: the complete context cannot enter the only lane on which the branch can change its cyclic order.

Corollary 5.2 does not prove that the map

$$a \mapsto \text{Break}_\Delta(a) \quad (5.13)$$

is too coarse to determine Safe-Hit. Such a quotient non-descent theorem requires two branches with equal break sets and different Safe-Hit predicates.

6 The Remaining Mobility Problem

The obstruction identifies the missing positive hypothesis. It is not enough that some lane be order breaking; one concrete five-token context must reach a usable breaking placement.

Definition 6.1 (mobility to a breaking witness).

Write

$$\text{MobToBreak}_a(x) \quad (6.1)$$

if an actual guarded identity-branch path carries x to a concrete state y in a lane C such that:

1. the marked pair of y is nonadjacent in C ;
2. $a|_C$ sends that concrete pair to an adjacent pair;
3. the same state $a_{\#}y$ admits the declared continuation to the target.

Proposition 6.2 (mobility-to-break sufficiency).

For every branch a ,

$$\text{MobToBreak}_a(x) \implies \text{SafeHit}_a(x). \quad (6.2)$$

Proof. Concatenate the guarded identity path in Definition 6.1, its pathwise branch- a simulation inherited from Paper XXXII, the globally enabled label-zero $a_{\#}$ step at y , and the declared continuation from the same state $a_{\#}y$. This branch step is not an additional guarded ordinary return. Every clause uses the witnesses supplied by the definition.

The proposition is intentionally conditional. The substantive open problem is to characterize when the cyclic phase-incidence fibers of Corollary 4.1 contain a usable breaking witness.

6.1 Open Problem 6.3 (minimal stabilizer incidence)

Find a quotient of the phase-incidence data that:

1. determines the return guard;
2. retains the concrete intermediate witness needed by Proposition 3.1;
3. detects mobility to a usable breaking placement; and
4. is strictly smaller than the raw five-token state relation.

Candidate summaries include the break set, the lane permutation, restriction cycle types, and relative phase data. None is promoted here as sufficient.

7 Computational Artifacts

The paper-owned evidence package is available under [experiments/paper33/](#) in the RIME repository. It has no dependency on the broader exploratory source tree.

Surface	Paper-owned path	Role
Bounded finite controls	<code>stabilizer_hostile_audit.py</code> and <code>results/</code>	Replays the retained full-stabilizer and capacity-isolation domains
Partial formalization	<code>lean/</code>	Checks the abstract capacity-obstruction implication spine
Development validation	<code>development-manifest.json</code> and <code>validation/</code>	Binds the 19-artifact paper-owned evidence closure
Release validation	<code>release-manifest.json</code> and the public receipt under <code>results/</code>	Binds the manuscript, reader PDF, bibliography, figure, nested development closure, environment, and validators

For all eight multi-lane domains with $6 \leq n \leq 9$, the producer exhausts 656 full stabilizer branches and verifies 656 instances of Proposition 3.1. No failure of (5.3) occurs in that bounded range.

For

$$(n, \Delta) = (10, 2), \quad (7.1)$$

the complete 2,880-branch scan records

$$80 \text{ ADJ}, \quad 160 \text{ INTERMEDIATE}, \quad 2640 \text{ SAME}. \quad (7.2)$$

The first intermediate branch is the $g = 2$ instance of Theorem 5.1. Its state counts are

$$|\text{Adj}| = 505, \quad |\text{SafeHit}| = 555, \quad |\text{SameLane}| = 560. \quad (7.3)$$

The package also checks the forward-invariant family used in Theorem 5.1 for $2 \leq g \leq 12$.

These calculations are bounded consistency controls. They are not proofs of Proposition 3.1 or Theorem 5.1, and the absence of a retained matched hostile pair is not a quotient-descent result.

A paper-owned Lean development machine-checks the data-independent capacity-obstruction spine of Theorem 5.1: a rank-five output cannot occupy a capacity-four lane; the declared ordinary-lane nonedge family is preserved by one step and hence by finite reachability; and that family is disjoint from the edge target. The formalization consumes, rather than constructs, the concrete branch family, cyclic lane arithmetic, lane-shape dichotomy, and ordinary-lane nonedge preservation proved above. It does not formalize the finite databases or promote them to all- g evidence.

The public receipt records performed replay and local closure verification, is excluded from its own closure, and is not an independent mathematical validation.

8 Claim Status and Boundary

Claim	Status	Scope
Exact source-addressed stabilizer descent	Proposition 3.1	Every branch permutation and pair of universal-return orbits
Cyclic phase parametrization	Corollary 4.1	Every source, intermediate, and target orbit in the declared cyclic action
Capacity-isolated break obstruction	Theorem 5.1	The family $n = 5g, \Delta = g$, every integer $g \geq 2$
Break nonemptiness is insufficient	Corollary 5.2	Full lane-partition stabilizer in the same all- g family
Mobility-to-break sufficiency	Proposition 6.2	Conditional on one concrete same-witness guarded path
Capacity-obstruction implication spine	Machine-checked partial Lean formalization	Abstract finite reachability system under the declared invariance and target-disjointness hypotheses
Retained finite scans	Bounded consistency control	All multi-lane domains $6 \leq n \leq 9$, the complete $(10, 2)$ domain, and $2 \leq g \leq 12$ family controls
Break-set quotient non-descent	Open	Requires matched branches with equal retained break data and different Safe-Hit predicates
Minimal sufficient incidence quotient	Open	Must preserve guards, same-witness composition, and usable breaking placement
General positive mobility-to-break theorem	Open	No all-stabilizer mobility criterion is claimed
Complete survivor incidence	Not claimed	Marked-pair return does not determine all spectator placements
Raw-to-typed transfer bridge	Not claimed	Raw reachability supplies no typed authority
Projectable-origin supply	Not claimed	Outside the intrinsic raw system
Recursive return and credit settlement	Not claimed	No reset-bound consequence

The finite controls do not enlarge the theorem domain. Raw reachability does not supply typed authority.

9 Conclusion

The full lane stabilizer changes the nature of return descent. A normalizer branch acts deterministically on universal-return orbits; a general stabilizer branch does not. The exact replacement is source-addressed orbit incidence, and its cyclic phase parametrization preserves the intermediate witness needed by the next partial return.

In this sense the full stabilizer is a natural endpoint of lane-only descent: the lane partition remains visible, but it no longer determines which concrete phase witness can pass the next guard.

Order breaking alone is nevertheless insufficient. In the family $n = 5g, \Delta = g$, the only breaking lane has capacity four while the candidate context occupies five points. Complete ordinary-lane contexts cannot enter the breaking lane, so same-lane nonedges survive as a forward-invariant obstruction.

The remaining classification problem is therefore no longer whether a branch breaks cyclic order somewhere. It is whether one concrete guarded context can reach a breaking placement and continue from that same witness.

References

- [1] Mikhail V. Berlinkov and Cyril Nicaud. Synchronizing almost-group automata. *International Journal of Foundations of Computer Science*, 31(8):1091–1112, 2020.
- [2] David Fernando Casas Torres. Completely reachable almost group automata, 2024.
- [3] Ján Černý. Poznámka k homogénnym experimentom s konečnými automatmi. *Matematicko-fyzikálny časopis Slovenskej Akadémie Vied*, 14:208–216, 1964.
- [4] WuJun Chen. Branch-normalized partial returns in single-defect circular automata. RIME Paper XXXI, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23050809>.
- [5] WuJun Chen. Cyclic lineage dynamics and rank-five first-exit frontiers. RIME Paper XXIX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23028361>.
- [6] WuJun Chen. Five-token dynamics beyond lane-order symmetry. RIME Paper XXXII, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23053169>.
- [7] WuJun Chen. Universal return groups and punctured rotations. RIME Paper XXX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23034428>.
- [8] Henk Don. The Černý conjecture and 1-contracting automata. *The Electronic Journal of Combinatorics*, 23(3):P3.12, 2016.
- [9] Lucien Dubuc. Sur les automates circulaires et la conjecture de Černý. *RAIRO Informatique théorique et applications*, 32:21–34, 1998.
- [10] David Eppstein. Reset sequences for monotonic automata. *SIAM Journal on Computing*, 19(3):500–510, 1990.
- [11] Robert Ferens and Marek Szykuła. Recognizing completely reachable automata in quadratic time. *ACM Transactions on Algorithms*, 22(2):1–36, 2026.
- [12] François Gonze and Raphaël M. Jungers. On completely reachable automata and subset reachability. In *Developments in Language Theory*, volume 11088 of *Lecture Notes in Computer Science*, pages 330–341. Springer, 2018.
- [13] Andrew Ryzhikov and Anton Shemyakov. Subset synchronization in monotonic automata. *Fundamenta Informaticae*, 162(2–3):205–221, 2018.
- [14] Mikhail V. Volkov. Synchronizing automata and the Černý conjecture. In *Language and Automata Theory and Applications*, volume 5196 of *Lecture Notes in Computer Science*, pages 11–27. Springer, 2008.
- [15] Mikhail V. Volkov. Synchronization of finite automata. *Russian Mathematical Surveys*, 77(5):819–891, 2022.
- [16] Petra Wolf. Synchronization under dynamic constraints. In *40th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2020)*, volume 182 of *Leibniz International Proceedings in Informatics*, pages 58:1–58:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020.